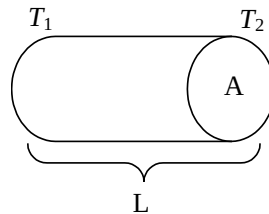


3. Fourier's law

(1) Fourier's law for heat diffusion

$$\boxed{q = \dot{E} = \dot{Q} = -kA \frac{dT}{dx}} \quad \text{In 3-d, } \mathbf{q}'' = -k\nabla T = -k \left(\frac{\partial T}{\partial x} \mathbf{i} + \frac{\partial T}{\partial y} \mathbf{j} + \frac{\partial T}{\partial z} \mathbf{k} \right)$$

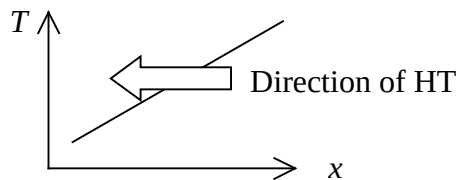


$$q = \dot{Q} = Ak \frac{\Delta T}{\Delta x} = Ak \frac{|T_1 - T_2|}{L}$$

Heat transfer rate $\sim$ (cross sectional area) $\times$ (Temperature difference) / (distance)

Note:

- HT rate is proportional to the "temperature gradient" ($\cong$ temperature difference/distance), NOT to temperature difference
- Minus sign ($-$) denotes the direction of heat flow.



- HT rate is a vector that has a direction from high T to low T .
Direction of HT $\perp$ Isothermal surfaces
- Thermal conductivity is a material property and a function of thermodynamic state (T , P , etc.). Particularly, temperature dependence of thermal conductivity has to be considered when a large temperature range is involved
- In general, thermal conductivity depends on the direction. If the material is isotropic, k is independent of direction and a constant. Otherwise, second-order tensor k_{ij}

(2) Microscopic view of heat transfer

- Energy carriers
 - Phonon: quantum of lattice vibration energy
 - Particles (electron, molecule, atom)
 - Photon: quantum of EM energy

- Characteristic length scales (mean free path)
 - The mean free path of momentum, energy, and mass carriers is the average distance that a carrier can travel without collision. It differentiates the micro-transport regime from the macroscale regime.
 - MFP: average distance a particle travels without a collision
 - $MFP = l = \frac{1}{\pi d^2 n}$ (d : particle diameter, n : number density)

(3) Thermal conductivity

- In general, $k_{sol} > k_{liq} > k_{gas}$

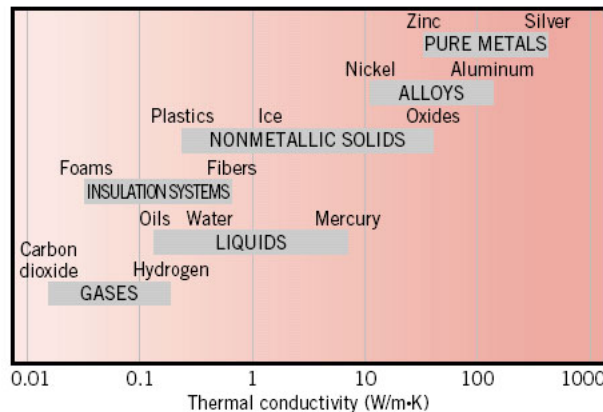


FIGURE 2.4 Range of thermal conductivity for various states of matter at normal temperatures and pressure.

- Solid: $k = (k \text{ by electron motion}) + (k \text{ by lattice vibration})$
 - Good electrical conductors are good thermal conductors as well (Wiedemann-Franz law)
 - Well-order materials (crystals) are good thermal conductors in comparison with amorphous materials (grain boundary scattering)
 - Size effect occurs when a film thickness is comparable to the MFP (boundary scattering)
 - Theoretical basis: solid state physics
- Gas: $k = \frac{1}{3} \rho v l C_v$
 - In general, $k_{gas} \uparrow$ as $T \uparrow$ because $v \uparrow$ as $T \uparrow$
 - Insensitive to P and ρ . ($n \uparrow$ as P , $\rho \uparrow$ but $l \downarrow$ as P , $\rho \uparrow$)

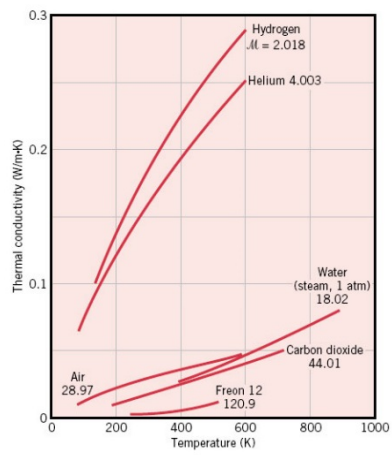


FIGURE 2.8 The temperature dependence of the thermal conductivity of selected gases at normal pressures. Molecular weights (M) of the gases are also shown.

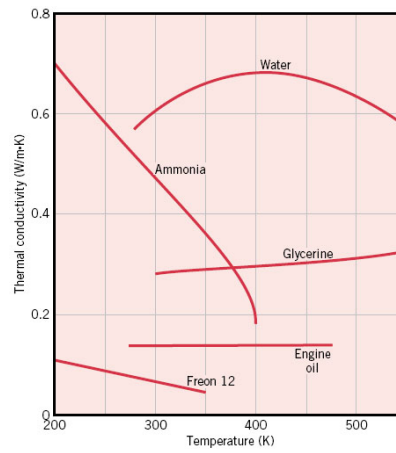


FIGURE 2.9 The temperature dependence of the thermal conductivity of selected nonmetallic liquids under saturated conditions.

- Thermal diffusivity: $\alpha \equiv \frac{k}{\rho c_p} = (\text{thermal conductivity})/(\text{density} \times \text{specific heat})$